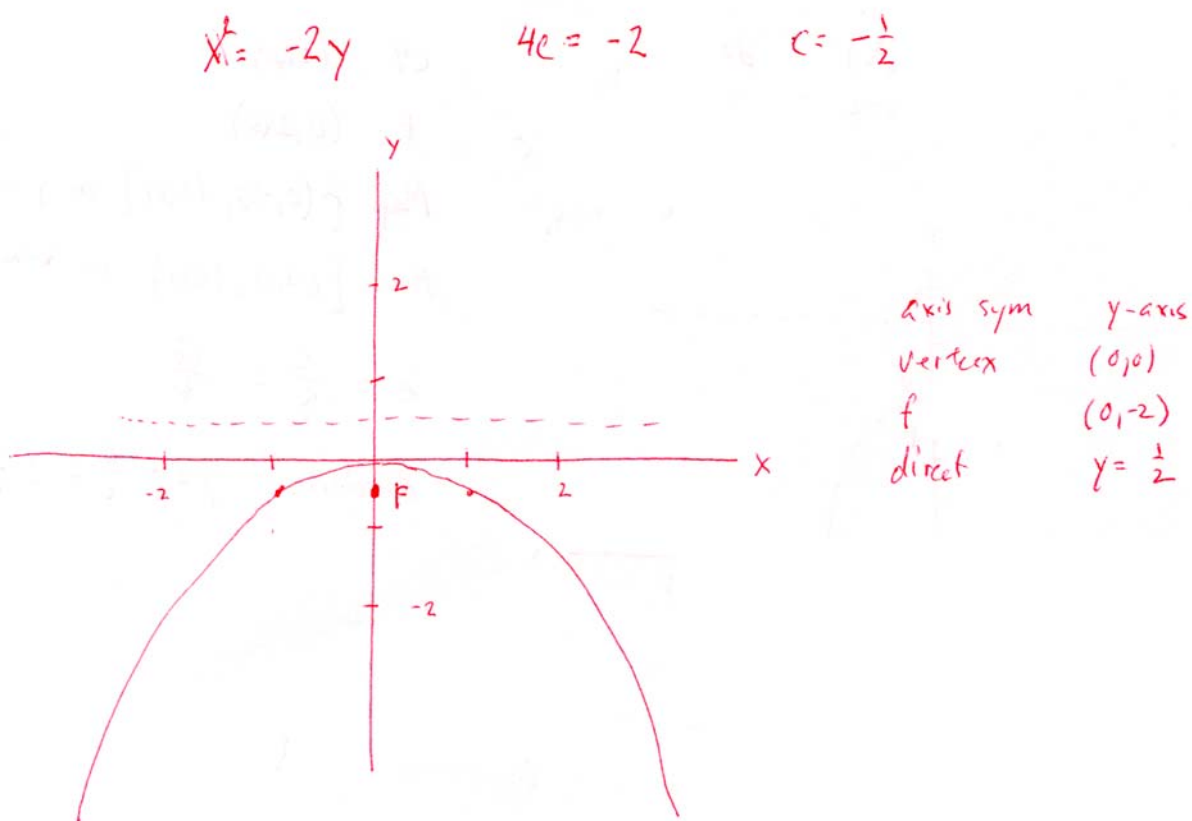


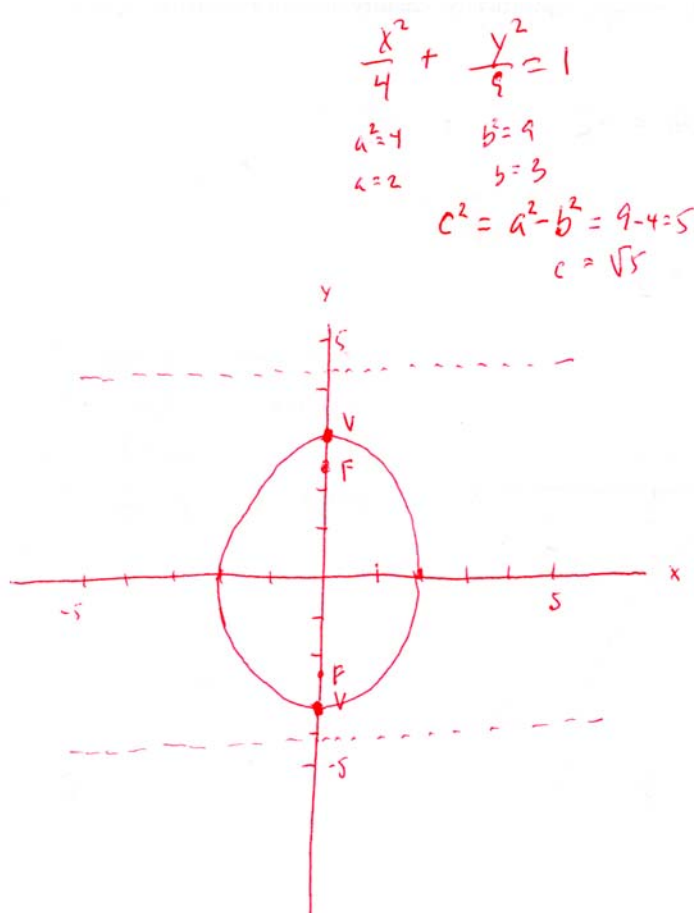
Name _____

Math 1350, Quiz #5

1. (Section 5.2, Problem 9) Sketch and discuss the graph of $x^2 = -2y$. Specifically, identify: axis of symmetry, vertex, focus, and directrix.



2. (Section 5.3, Problem 7) Sketch and discuss the graph of $9x^2 + 4y^2 = 36$. Specifically, identify: vertices, covertices, major and minor axes, foci, eccentricity, and directrices.



$$V (0, \pm 3)$$

$$CV (\pm 2, 0)$$

$$F (0, \pm \sqrt{5})$$

$$\text{Major } [(0, -3), (0, 3)] \text{ on } y\text{-axis}$$

$$\text{Minor } [(-2, 0), (2, 0)] \text{ on } x\text{-axis}$$

$$e = \frac{c}{a} = \frac{\sqrt{5}}{3}$$

$$\text{directrices } y = \pm \frac{a}{e} = \pm \frac{9}{\sqrt{5}}$$